

DESIGN CODE OF CLA

/home/dhruv/cao_lab/vivado_proj/cao_lab_u24ev057/rtl/cla8.vhd [VHDL]

```
1  -----
2  --
3  -- Company:
4  -- Engineer:
5  --
6  -- Create Date: 08/13/2026 04:46:52 PM
7  -- Design Name:
8  -- Module Name: cla8 - structural
9  -- Project Name:
10 -- Target Devices:
11 -- Tool Versions:
12 -- Description:
13 --
14 -- Dependencies:
15 --
16 -- Revision:
17 -- Revision 0.01 - File Created
18 -- Additional Comments:
19 --
20 -----
21 --
22 library IEEE;
23 use IEEE.STD_LOGIC_1164.ALL;
24
25 -- Uncomment the following library declaration if using
26 -- arithmetic functions with Signed or Unsigned values
27 --use IEEE.NUMERIC_STD.ALL;
28
29 -- Uncomment the following library declaration if instantiating
30 -- any Xilinx leaf cells in this code.
31 --library UNISIM;
32 --use UNISIM.VComponents.all;
33
34 entity cla8 is
35     Port ( a : in STD_LOGIC_VECTOR (7 downto 0);
36           b : in STD_LOGIC_VECTOR (7 downto 0);
37           cin : in std_logic;
38           sum : out STD_LOGIC_VECTOR (7 downto 0);
39           cout : out STD_LOGIC);
40 end cla8;
41
42 architecture structural of cla8 is
43
44     signal c : std_logic_vector(7 downto 0);
45
46     signal p : std_logic_vector(7 downto 0);
```

47	signal g : std_logic_vector(7 downto 0);
48	
49	begin
50	
51	p <= a xor b;
52	g <= a and b;
53	
54	
55	c(0) <= cin;
56	c(1) <= g(0) or (p(0) and (cin));
57	c(2) <= g(1) or (p(1) and (g(0) or (p(0) and (cin))));
58	c(3) <= g(2) or (p(2) and (g(1) or (p(1) and (g(0) or (p(0) and (cin))))));
59	c(4) <= g(3) or (p(3) and (g(2) or (p(2) and (g(1) or (p(1) and (g(0) or (p(0) and (cin))))))));
60	c(5) <= g(4) or (p(4) and (g(3) or (p(3) and (g(2) or (p(2) and (g(1) or (p(1) and (g(0) or (p(0) and (cin))))))));
61	c(6) <= g(5) or (p(5) and (g(4) or (p(4) and (g(3) or (p(3) and (g(2) or (p(2) and (g(1) or (p(1) and (g(0) or (p(0) and (cin))))))));
62	c(7) <= g(6) or (p(6) and (g(5) or (p(5) and (g(4) or (p(4) and (g(3) or (p(3) and (g(2) or (p(2) and (g(1) or (p(1) and (g(0) or (p(0) and (cin))))))));
63	cout <= g(7) or (p(7) and (g(6) or (p(6) and (g(5) or (p(5) and (g(4) or (p(4) and (g(3) or (p(3) and (g(2) or (p(2) and (g(1) or (p(1) and (g(0) or (p(0) and (cin))))))));
64	
65	sum(0) <= a(0) xor b(0) xor c(0);
66	sum(1) <= a(1) xor b(1) xor (g(0) or (p(0) and (cin)));
67	sum(2) <= a(2) xor b(2) xor (g(1) or (p(1) and (g(0) or (p(0) and (cin))));
68	sum(3) <= a(3) xor b(3) xor (g(2) or (p(2) and (g(1) or (p(1) and (g(0) or (p(0) and (cin))))));
69	sum(4) <= a(4) xor b(4) xor (g(3) or (p(3) and (g(2) or (p(2) and (g(1) or (p(1) and (g(0) or (p(0) and (cin))))));
70	sum(5) <= a(5) xor b(5) xor (g(4) or (p(4) and (g(3) or (p(3) and (g(2) or (p(2) and (g(1) or (p(1) and (g(0) or (p(0) and (cin))))));
71	sum(6) <= a(6) xor b(6) xor (g(5) or (p(5) and (g(4) or (p(4) and (g(3) or (p(3) and (g(2) or (p(2) and (g(1) or (p(1) and (g(0) or (p(0) and (cin))))));
72	sum(7) <= a(7) xor b(7) xor (g(6) or (p(6) and (g(5) or (p(5) and (g(4) or (p(4) and (g(3) or (p(3) and (g(2) or (p(2) and (g(1) or (p(1) and (g(0) or (p(0) and (cin))))));
73	
74	end structural;

TEST BENCH CODE OF CLA

/home/dhruv/cao_lab/vivado_proj/cao_lab_u24ev057/tb/cla8_tb.vhd [VHDL]	
1	-----
	--
2	-- Company:
3	-- Engineer:
4	--
5	-- Create Date: 08/13/2026 05:08:58 PM
6	-- Design Name:
7	-- Module Name: cla8_tb - testing

```

8  -- Project Name:
9  -- Target Devices:
10 -- Tool Versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
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21
22 library IEEE;
23 use IEEE.STD_LOGIC_1164.ALL;
24
25 -- Uncomment the following library declaration if using
26 -- arithmetic functions with Signed or Unsigned values
27 --use IEEE.NUMERIC_STD.ALL;
28
29 -- Uncomment the following library declaration if instantiating
30 -- any Xilinx leaf cells in this code.
31 --library UNISIM;
32 --use UNISIM.VComponents.all;
33
34 entity cla8_tb is
35 -- Port ( );
36 end cla8_tb;
37
38 architecture testing of cla8_tb is
39
40     component cla8 is
41         Port ( a : in STD_LOGIC_VECTOR (7 downto 0);
42               b : in STD_LOGIC_VECTOR (7 downto 0);
43               cin : in std_logic;
44               sum : out STD_LOGIC_VECTOR (7 downto 0);
45               cout : out STD_LOGIC);
46     end component;
47
48
49     signal a : std_logic_vector(7 downto 0);
50     signal b : std_logic_vector(7 downto 0);
51     signal s : std_logic_vector(7 downto 0);
52     signal cin : std_logic;
53     signal cout : std_logic;
54
55
56
57 begin
58

```

59	
60	cla_test : cla8 port map(
61	a => a,
62	b => b,
63	cin => cin,
64	sum => s,
65	cout =>cout
66	);
67	
68	
69	process
70	
71	begin
72	cin <= '0';
73	a <= "01010101";
74	b <= "10101010";
75	
76	wait for 10ns;
77	
78	a <= "00110011";
79	b <= "11001100";
80	
81	wait for 10ns;
82	
83	a <= "10101010";
84	b <= "11001100";
85	
86	wait for 10ns;
87	
88	cin <= '1';
89	a <= "00110001";
90	b <= "10011100";
91	
92	wait for 10ns;
93	
94	a <= "11001010";
95	b <= "11000011";
96	
97	wait for 10ns;
98	
99	a <= "11111111";
100	b <= "11111111";
101	
102	wait for 10ns;
103	
104	cin <= '0';
105	a <= "00000000";
106	b <= "00000000";
107	
108	wait for 10ns;
109	

110	a <= "01011010";
111	b <= "11110011";
112	
113	wait for 10ns;
114	
115	cin <= '1';
116	a <= "00000001";
117	b <= "10000111";
118	
119	wait for 10ns;
120	
121	a <= "10101010";
122	b <= "11111111";
123	
124	
125	wait;
126	
127	end process;
128	
129	
130	
131	end testing;

SCREENSHOT OF CODE

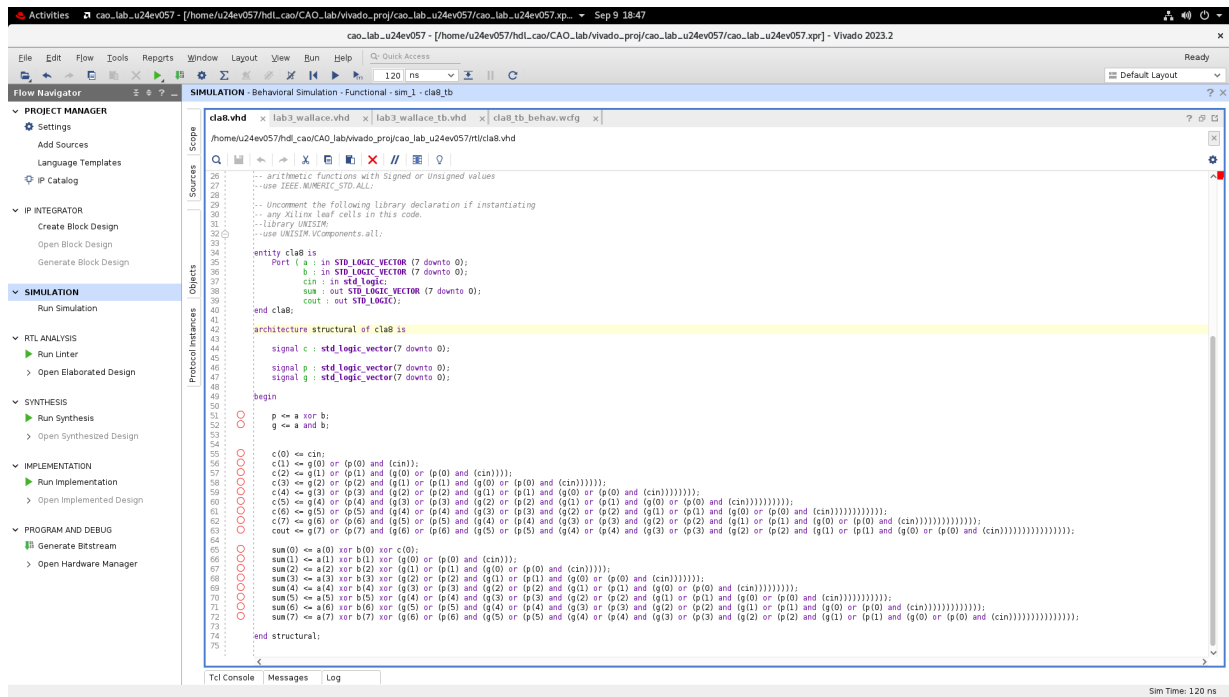


Figure 1: Design

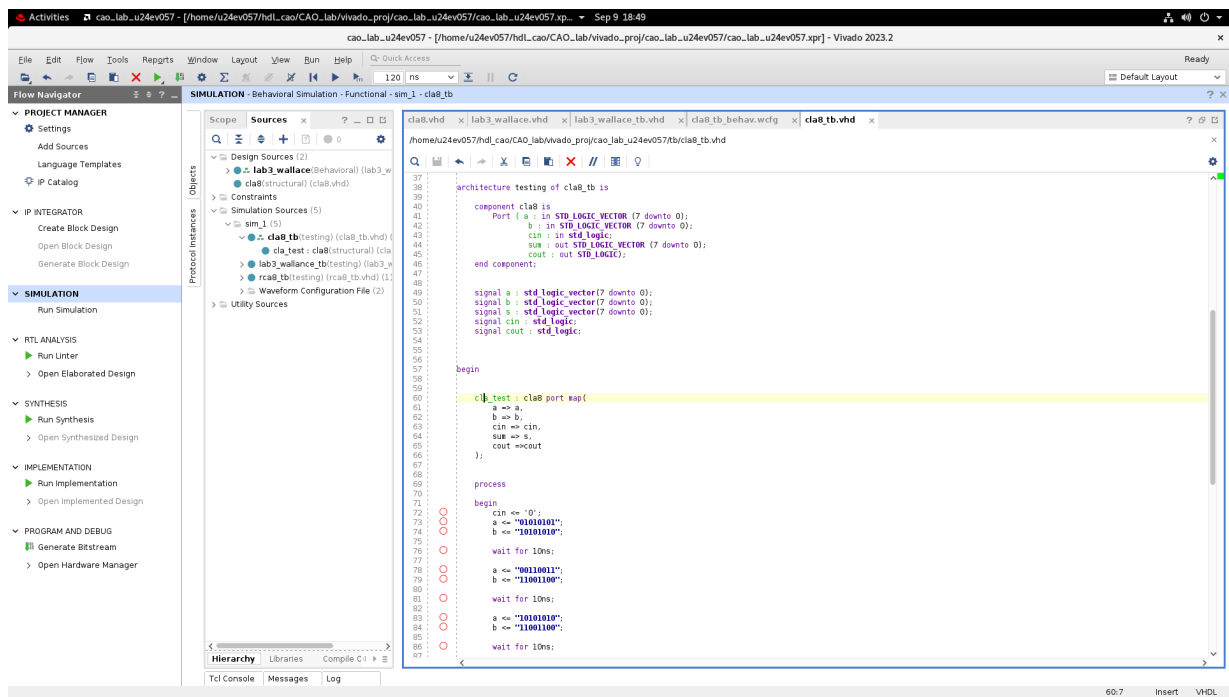


Figure 2: Tb

WAVEFORM

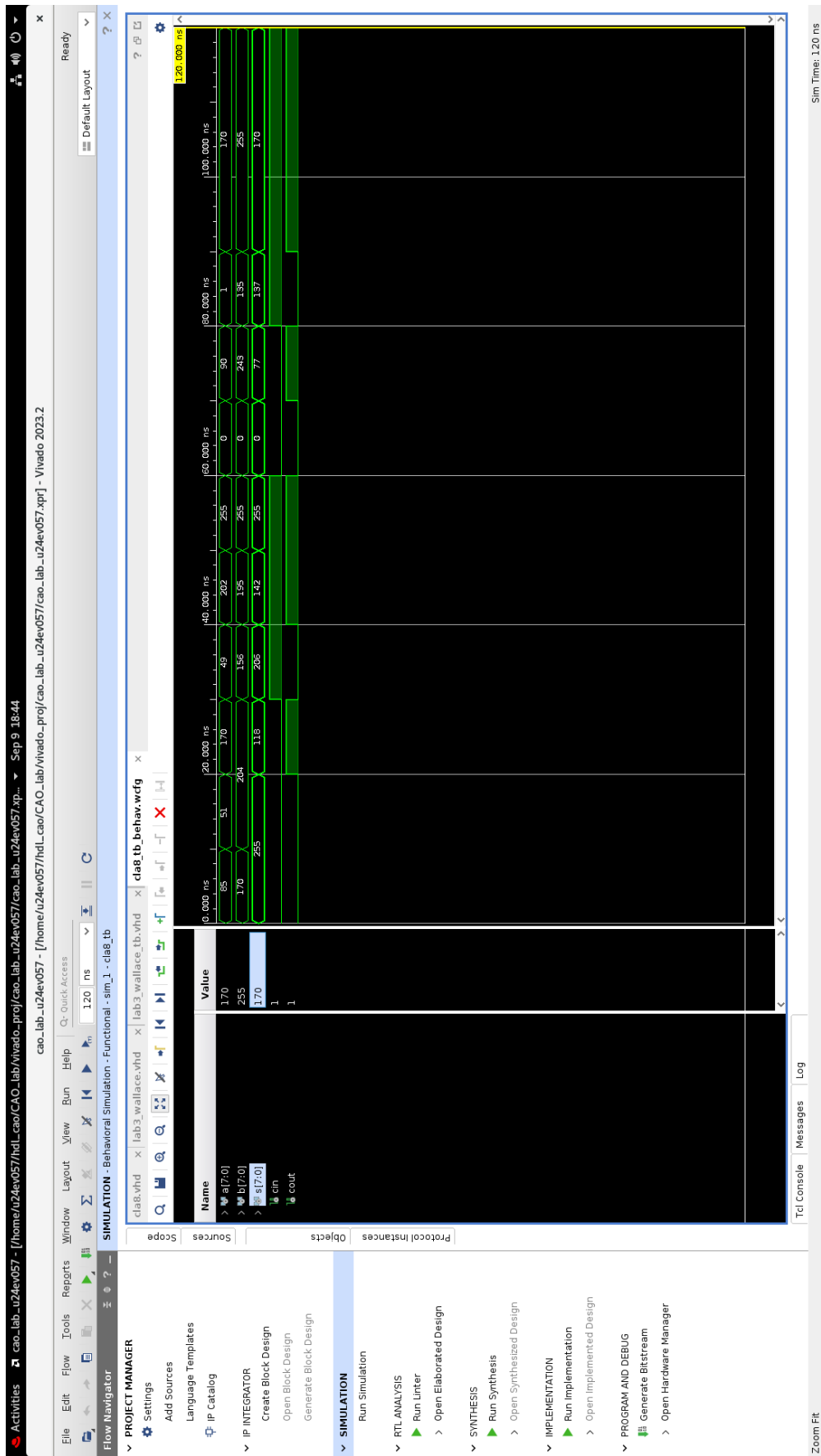


Figure 3: Waveform

